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A Project Report

On

“Smart sericulture: Automated disease diagnosis in silkworm using CNN”

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in

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Certified that the 7th Semester Project titled “**Smart sericulture: Automated disease diagnosis in silkworm using CNN**” is a Bonafide work carried out by **Harshitha N(4VM22CS013), Keerthana K T(4VM22CS014), Aruna P(4VM23CS402) and Sinchana Y M(4VM23CS412)** in partial fulfilment for the award of degree of Bachelor of Engineering in **Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi, during the year 2025-26. The Project report has been approved as it satisfies the academic requirements with respect to the project work prescribed for Bachelor of Engineering Degree.

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ABSTRACT

The productivity and sustainability of the sericulture industry largely depend on the health of silkworms and the quality of their host plants, particularly mulberry leaves. Traditional methods of disease detection in silkworms and mulberry plants are manual, time consuming, and often prone to human error. To address these challenges, this project aims to develop an intelligent, web-based system that leverages advanced image processing and deep learning techniques to automatically diagnose diseases affecting silkworms and mulberry leaves. By providing accurate and real-time analysis, the system supports farmers and researchers in maintaining optimal rearing conditions and improving overall yield in sericulture.

The proposed system integrates two diagnostic modules based on Convolutional Neural Networks (CNN). The first module classifies silkworm health conditions into five categories—Healthy, Grasserie, Flacherie, Muscardine, and Pebrine—using silkworm images. The second module identifies common foliar infections in mulberry leaves by analyzing leaf images. Both modules are deployed on a user-friendly web-based platform that allows users to upload images, receive instant disease predictions, view confidence levels, and access recommended management practices. The methodology involves image preprocessing, feature extraction using deep CNN architectures, model training and validation, and seamless integration of both modules into the web interface for real-time diagnostic support.

Keywords: Sericulture, Silkworm Diseases, Mulberry Leaf Infections, Convolutional Neural Networks (CNN), Deep Learning, Image Classification, Web Application, Smart Farming, Disease Detection, Precision Agriculture.

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Abbreviation

CNN	Convolutional Neural Network
ML	Machine Learning
DL	Deep Learning
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
RF	Random Forest
ROC	Receiver Operating Characteristic
AUC	Area-Under-Curve
XAI	Explainable AI / Explainability
Grad-CAM	Gradient-weighted Class Activation Mapping

CHAPTER 1

INTRODUCTION

Sericulture, the practice of rearing silkworms for silk production, is an important rural industry in countries like India, China, and Thailand. In India, especially in states such as Karnataka and Tamil Nadu, sericulture provides steady income to thousands of small farmers. The success of silk production largely depends on keeping silkworms healthy, as they are extremely sensitive to environmental changes and disease infections. However, common diseases such as Pebrine, Grasserie, Muscardine, and Flacherie spread quickly and can destroy an entire batch of larvae if not detected early. Since early symptoms are often subtle, farmers usually struggle to identify diseases on time, leading to heavy economic losses.



Fig 1: Silkworm

Traditional disease detection mainly relies on visual inspection and expert guidance. This method is slow, subjective, and not always accessible to all farmers. Convolutional Neural Networks (CNNs), which specialize in image classification, can learn to detect disease patterns in silkworm images with high accuracy. By training a CNN model using images of both healthy and infected larvae, it becomes possible to identify diseases automatically and quickly.

The project Smart Sericulture: Automated Disease Diagnosis in Silkworms Using CNN focuses on building such an intelligent system. The process involves collecting silkworm images, preprocessing them, augmenting the dataset, and training a CNN model using pretrained dataset. Once trained, the system can be deployed through a mobile or web application, allowing farmers to take a photo of larvae and receive insta diagnosis.

1.1 Problem statement

Manual detection of silkworm and mulberry leaf diseases is often slow, inaccurate, and heavily dependent on expert involvement, making large-scale monitoring challenging. The absence of automated tools for early diagnosis leads to delayed treatment, rapid disease spread, and significant loss in both silkworm rearing and mulberry cultivation. This project proposes a CNN-based model to automate the identification of diseases in silkworms and mulberry leaves through image classification, ensuring rapid, accurate, and reliable detection to support modern sericulture practices.

1.2 Aim & Objectives

Aim:

The aim of the Silkworm Disease Detection using Convolutional Neural Networks (CNN) is to develop an intelligent, automated system that can accurately identify and classify common silkworm diseases from image inputs. This system is designed to support sericulture by enabling early diagnosis, reducing the dependence on human experts, minimizing silkworm mortality, and ensuring healthier silkworm cultivation through rapid and reliable disease detection.

Objectives:

1.2.1. To develop an automated system using CNN for detecting diseases in both silkworms and mulberry leaves.

This objective focuses on creating an intelligent system that can automatically analyze images of silkworms and mulberry leaves using Convolutional Neural Networks (CNN). The system will study visual patterns, identify abnormalities, and detect the presence of various diseases without any manual inspection. This automation improves the speed and accuracy of the disease detection process in sericulture.

1.2.2. To identify infections at an early stage with high accuracy, helping prevent crop and cocoon loss.

Early detection is crucial in sericulture because infections spread quickly and can cause major damage. This objective aims to ensure that the system is capable of recognizing diseases at their initial stage, allowing farmers to take timely preventive measures. High-accuracy detection helps in reducing mortality of silkworms and minimizing damage to mulberry leaves, ultimately preventing financial loss.

1.2.3. To reduce manual effort and time required for disease diagnosis in sericulture.

Traditionally, farmers or experts must visually inspect silkworms and leaves, which is slow, labor-intensive, and prone to human error. With this objective, the project aims to automate the entire diagnosis process. By using image processing and AI, the system significantly reduces manual work and saves valuable time, especially during large-scale farming.

1.2.4. To provide farmers with a user-friendly and scalable tool for timely preventive measures.

The system should be simple to operate, even for farmers with minimal technical knowledge. A user-friendly interface, easy image uploading, and clear disease results help farmers take quick action. The tool will also be scalable, meaning new diseases, additional features, and more data can be incorporated easily in the future as the system evolves.

1.2.5. To store and maintain a disease database for future predictions and improved accuracy.

This objective ensures the creation of a centralized database containing images, disease results, and classification data. This stored information helps the model learn from past cases, improving its prediction accuracy over time. A well-maintained database is also useful for future research, trend analysis, and the introduction of advanced machine learning techniques.

1.3 Methodology

The methodology of the project involves a systematic process to develop an automated system capable of identifying silkworm diseases using Convolutional Neural Networks (CNNs). The workflow includes data collection, preprocessing, model design, training, evaluation, and deployment. Each step ensures that the system performs accurately and consistently in real- world sericulture environments.

1.3.1 Data Collection:

Images of healthy and diseased silkworms are collected from sericulture farms, research centers, and online sources. The dataset includes multiple disease types such as Grasserie, Pebrine, Flacherie, and Muscardine. High-quality images captured under different lighting and angles help improve the robustness of the model.

1.3.2 Data Preprocessing:

Collected images undergo preprocessing to prepare them for model training. Preprocessing ensures uniformity and improves model performance.

This includes:

- Resizing images to a fixed dimension (e.g., 224×224 pixels)
- Normalizing pixel values
- Converting to RGB format
- Data augmentation (rotation, flipping, zooming) to increase dataset size and reduce overfitting

1.3.3 Dataset Splitting:

The dataset is divided into:

- Training set (70%)
- Validation set (20%)
- Testing set (10%)

1.3.4 CNN Model Design:

A CNN architecture (custom or pre-trained like VGG16/ResNet/MobileNet) is selected. The model includes:

- Convolution layers for feature extraction
- Pooling layers for dimensionality reduction

- Fully connected layers for classification
- Soft-max output layer for predicting disease classes

1.3.5 Model Training:

The training process involves feeding the preprocessed images into the CNN. Backpropagation and optimization algorithms (like Adam) update the weights to minimize loss. The model learns disease features such as texture changes, spots, fungal growth, or discoloration.

1.3.6 Model Evaluation:

Performance is tested using metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

This ensures reliability before real-world deployment.

1.3.7 Deployment:

The trained model is deployed through a simple web application. Farmers can capture silkworm images using a smartphone and receive instant disease predictions. The interface is made user-friendly and suitable for rural users.

1.4 Scope of the project

1. Project Objective

The primary goal of this project is to assist sericulture (silk farming) farmers by leveraging technology to detect diseases early and provide market information. It aims to bridge the gap between traditional farming and modern AI-based diagnostics.

2. Target Audience

- Sericulture Farmers: specifically those who may not have access to expert agriculturalists for immediate disease diagnosis.
- Region: Likely Karnataka/India, given the inclusion of Kannada language support and scraping of the Karnataka Sericulture Department website.

3. Functional Scope

The project currently covers three main domains:

A. Disease Detection (AI/ML)

➤ Silkworm Disease Diagnosis:

- Users can upload images of silkworms.
- The system uses a Deep Learning model (model1.h5) to classify the image into categories: Flacherie, Grasserie, Muscardine, Pebrine, or Healthy.

➤ Mulberry Leaf Disease Diagnosis:

Users can upload images of mulberry leaves (the primary food for silkworms). The system uses a Deep Learning model (model2.h5) to detect issues like: Leaf Rust, Leaf Spot, Stem Canker, Powdery Mildew, and Root Knot.

Remedies: Provides control measures and treatment advice in both English and Kannada for detected diseases.

B. Market Intelligence

Cocoon Market Prices: Includes a web scraper (price.py) that automatically fetches daily commercial cocoon rates from the Karnataka Sericulture government website.

Downloads and displays the latest price lists (PDFs) to help farmers decide when and where to sell.

C. Knowledge Portal

Farming Guide:

- Provides week-by-week guidance for silkworm rearing (Week 1 to Week 5).
- Educational content is available in both English and Kannada to ensure accessibility for local farmers.

D. Technical Scope

- a. Web Framework: Flask (Python) for the backend and routing.
- b. Frontend: HTML/CSS (Templates) for the user interface.
- c. AI/ML: TensorFlow/Keras for image classification models.
- d. Automation: BeautifulSoup4 for web scraping government data.

E. Future Scope

Mobile Application: Converting this web app into a mobile app for easier field usage. Real-time Alerts: SMS/WhatsApp notifications for market prices or disease outbreaks. Community Forum: A space for farmers to share issues and solutions.

1.5 Limitations

1. Critical Security Vulnerability: Unsafe File Uploads

- Consequence: If the server has permissions, this could overwrite critical system files, upload a virus to a sensitive folder, or replace your main1.py with a malicious script that steals data.

2. Performance & Scalability: Global Model Loading

- Memory Hog: Both deep learning models are loaded into RAM immediately when the server starts, even if no one ever uses the prediction feature. This can crash smaller servers (like a free tier cloud instance).
- Slow Startup: The application will take a long time to start up because it has to load these heavy files first.
- Concurrency: TensorFlow Graph objects are not always thread-safe by default in Flask. If two users try to predict at the exact same time, the server might crash or return errors.

3. Fragile Logic: "String Matching" for Predictions

- Unreliable: This logic assumes that if a probability is very small (e.g., $1.23e-05$), it's an "error" or "healthy". This is mathematically incorrect. A model outputting a low probability just means it's unsure, or it's a valid low probability for that class.
- Bug Prone: If the model improves and outputs 0.00001 instead of $1e-05$, your code behavior changes completely without you touching it.

4. Maintainability: Hardcoded Paths & Secrets

- Portability: If you move this code to a different folder or a different computer (like a Linux server), it might break because it can't find the files.
- Updates: If the URL for the Karnataka Sericulture website changes, you have to hunt through your code to find and fix it. These should be in a separate configuration file.

5. Code Quality: Dead Code.

Contains large blocks of commented-out code (lines 1-39, 40-79, 80-120, etc.). Why it's a problem: It makes the file hard to read and confusing for other developers (or you in the future) to know which part is actually running. It increases the file size unnecessarily.

1.6 Organization of the report

The report on "Smart Sericulture: Automated Disease Diagnosis in Silkworm Using CNN" is structured systematically to provide a comprehensive understanding of the project, covering all aspects from conceptualization to experimental analysis. The report is divided into seven main chapters along with supporting sections such as references, appendices, and acknowledgments.

Chapter 1: Introduction

This chapter introduces the background and significance of sericulture, emphasizing the economic and ecological importance of silkworm rearing. It explains the challenges faced due to silkworm diseases and the necessity of automated disease detection systems. The objectives and scope of the project are clearly defined, along with the motivation behind integrating Convolutional Neural Networks (CNN) for disease diagnosis. Key terminologies, definitions, and the overall research problem are also presented to provide context for the reader.

Chapter 2: Literature Review

This chapter reviews previous research and existing techniques related to silkworm disease detection and image-based analysis. It highlights various methods of disease identification, including traditional manual inspection and computer vision-based approaches. Comparisons of different machine learning and deep learning algorithms, particularly CNNs, are discussed, establishing the need for the proposed automated system. The literature review concludes by identifying gaps in current methods that the project aims to address.

Chapter 3: Methodology

This chapter outlines the detailed methodology adopted for the project. It covers data collection, including sources of silkworm images and disease samples. Preprocessing steps, such as image resizing, normalization, and augmentation, are explained.

The CNN architecture, including layers, activation functions, and training procedures, is described comprehensively. This chapter also explains the training, validation, and testing process, along with hyperparameter selection and optimization techniques.

Chapter 4: System Design and Implementation

This chapter presents the design of the automated disease diagnosis system. It includes both hardware (if any) and software components, system architecture diagrams, and workflow processes. The implementation of CNN models using frameworks such as TensorFlow or PyTorch is detailed. The chapter also illustrates how the system integrates input data, processes it, and generates diagnostic output.

Chapter 5: Results and Discussion

In this chapter, the results obtained from the CNN model are presented using tables, graphs, and confusion matrices. Model performance metrics such as accuracy, precision, recall, and F1-score are analyzed. Comparative discussions on the effectiveness of the model with existing methods are included. The chapter also discusses possible reasons for misclassifications and limitations observed during testing.

Chapter 6: Conclusion and Future Work

This chapter summarizes the achievements of the project, emphasizing its contribution to smart sericulture. It outlines how the automated system enhances disease diagnosis efficiency and reduces human intervention. Suggestions for future improvements, such as expanding the dataset, integrating IoT-based monitoring, or using more advanced deep learning models, are also presented.

References

All cited sources, research papers, books, and online references are listed in a standard citation format.

CHAPTER 2

LITERATURE SURVEY

1. A Novel Approach for Silkworm Disease Detection and Classification By Using CNN and Image Processing Techniques Shilpa B Kodli, Anita (2025)

A machine learning-based system utilizing Convolutional Neural Networks (CNN) and image processing is proposed to identify silkworm diseases such as Flacherie, Grasserie, and Muscardine. The system accurately classifies silkworm images into healthy or diseased categories, achieving 99% accuracy. By leveraging deep learning techniques, this automated system provides timely disease detection and treatment recommendations, potentially reducing economic losses in the silk industry and improving sericulture practices.

2. Silkworm yield prediction: Exploring Leaf types and Species Suchitha H S, Anish A, Gagandeep J E, Manoj M, Manu G (2025)

This study compares the performance of Random Forest and Linear Regression machine learning models in predicting silkworm yield based on leaf types and species characteristics. The Random Forest model, with its ensemble learning capabilities, effectively captures complex relationships between predictors and silkworm yield. The comparative analysis highlights the strengths of each approach, providing insights into their predictive capabilities. The study's results contribute to optimizing sericulture practices by offering a predictive framework for silkworm yield.

3. A comprehensive prediction system for silkworm acute toxicity assessment of environmental and in-silico pesticides Yutong Liu, Hou hu Zhang, Yue Yu, Hua Wu ,Bing Wu, Jieshu Qian (2024)

This study developed a machine learning approach to predict pesticide toxicity on silkworms, creating a comprehensive voting model that combines Bayesian Ridge, K Neighbors Regressor, and Random Forest Regressor algorithms. The model accurately predicts lethal concentration 50% (LC50) values and provides confidence intervals and structural alerts analysis. By leveraging ML, the study enables comprehensive evaluation of ecological toxicity risk, designs new low-toxic pesticide.

4. Grasserie disease detection in silkworm using machine learning methods Manju G (2024)

A machine learning approach was developed to predict pesticide toxicity on silkworms, utilizing a comprehensive voting model that combines multiple regression algorithms to accurately predict lethal concentration 50% (LC50) values. This model provides confidence intervals and structural alerts analysis, enabling comprehensive evaluation of ecological toxicity risk. The study also designed new low-toxic pesticide alternatives and developed software for environmental risk assessment, offering a cost-effective solution for predicting toxicity and accelerating risk assessment.

5. Grasserie disease Identification in Bombyx Mori Silkworm using Ensemble Learning Approach Ensemble Learning Approach Raghavendra C G, Dharini Raghavan (2022)

A machine learning-based system is being developed to detect diseases in silkworms, crucial for sericulture. The approach involves image processing and classification techniques, utilizing Local Binary Patterns (LBP) for feature extraction and Principal Component Analysis (PCA) for dimensionality reduction. An ensemble learning approach or decision tree classifier can be used to identify Grasserie disease in Bombyx mori silkworms accurately, achieving high accuracy (92.1%) and promising performance metrics. The developed algorithm enables machines to make decisions and can be implemented on hardware for real-time applications, potentially reducing economic losses in sericulture.

6. Development of a Muga Disease Early Warning System – A Mobile-Based Service for Seri Farmers Jonali Goswami, Nasreen Rasid, D K Gogoi (2021)

A Muga disease early warning system was developed using data analysis to predict Flacherie infestation in silkworms. The system uses relationships between rearing/production data, land use/land cover, temperature, and meteorological parameters to forecast disease outbreaks 5-10 days in advance.

This enables farmers to take precautionary measures, potentially reducing crop losses. A mobile-based application and dashboard were created for easy access and timely decision-making. A machine learning approach is used to detect Grasserie disease in silkworms through image processing and classification. The model utilizes Local Binary Patterns (LBP) for feature extraction, Principal Component Analysis (PCA) for dimensionality reduction, and a decision tree classifier or ensemble learning approach to achieve high accuracy (92.1%) in disease detection.

7. The evolutionary road from wild moth to domestic silkworm Hui Xiang, Xiaojing Liu, Muwang Li (2018)

A web-based application and dashboard were developed for a Muga disease early warning system, utilizing data analysis and computational techniques to predict Flacherie infestation in silkworms. The system leverages relationships between environmental factors and disease outbreaks to forecast outbreaks 5-10 days in advance, enabling farmers to take timely action.

8. Mulberry leaves diseases and disease identification techniques, Sandeep Kumar S and Usha Divarkarla

It explains how visual symptoms such as discoloration and leaf-spot patterns can be analyzed using segmentation, feature extraction, and classification methods. The study highlights the importance of early, automated detection to support mulberry farming and silkworm health. It also points out that more standardized datasets and detailed experimental studies are needed to improve accuracy and practical adoption.

9. Classification of healthy and diseased mulberry leaves using machine learning, IEEE, 2024. Shilpashree P. S., Sampath G. D., Pavankumaraswamy H. S., Sagar B. E., and Chaithra S K

The paper proposes a machine learning approach to classify mulberry leaves as healthy or diseased using image data. The authors preprocess the images, extract key features, and evaluate different ML models for accurate classification. Their system shows good performance in identifying leaf diseases early. This helps improve silkworm health and supports better cocoon production.

CHAPTER 3

METHODOLOGY

The methodology for developing the Smart Sericulture: Automated Disease Diagnosis in Silkworm Using CNN project is designed to ensure accurate, automated, and efficient identification of silkworm diseases. The approach follows a structured workflow involving data collection, preprocessing, CNN model development, training, evaluation, and deployment. Each stage is critical for ensuring reliable results and practical usability in real-world sericulture environments.

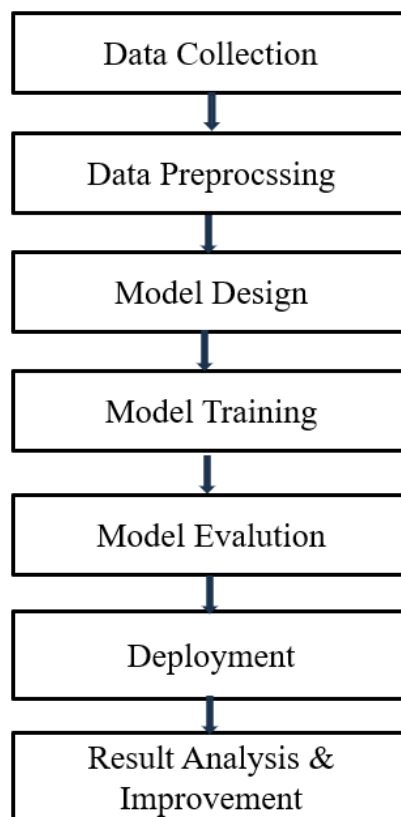


Fig 3.1: System methodology

1. Data Collection

The first step involves gathering images of healthy and diseased silkworms from multiple sources, including sericulture farms, research centers, and publicly available datasets. The collected dataset contains images of key silkworm diseases such as Grasserie, Pebrine, Flacherie, and Muscardine.

Care is taken to include images under varied lighting conditions, angles, and resolutions to make the dataset robust. High-quality and diverse images improve the CNN's ability to generalize and recognize diseases in different environmental conditions.

2. Data Preprocessing

Before feeding the images to the CNN, preprocessing is performed to standardize the data and enhance model performance. Key preprocessing steps include:

- Resizing all images to a fixed dimension (e.g., 224×224 pixels) for uniform input.
- Normalization of pixel values to a range [0, 1] to improve convergence during training.
- Conversion to RGB format to maintain color consistency.
- Data Augmentation such as rotations, flipping, zooming, and brightness adjustments to artificially expand the dataset and reduce overfitting.

These steps ensure that the CNN can efficiently learn patterns from the images, regardless of variations in the input data.

3. Dataset Splitting

To facilitate effective training and evaluation, the dataset is divided into three subsets:

- Training set (70%)– used to train the CNN model.
- Validation set (20%) – used to fine-tune hyperparameters and prevent overfitting.
- Testing set (10%) – used to evaluate the final model’s performance objectively.

This splitting ensures that the model is robust and can generalize well to unseen data.

4. CNN Model Design.

A Convolutional Neural Network (CNN) is employed for automatic feature extraction and disease classification. The architecture consists of:

- Convolutional layers – to detect and extract disease-relevant features such as spots.
- Pooling layers – to reduce dimensionality and computation while retaining key features.
- Fully connected layers – to interpret features and map them to disease classes.

5. Model Training

During training, the preprocessed images are fed into the CNN. The model learns by adjusting weights through backpropagation and optimization algorithms. The loss function measures the prediction error, guiding the model to improve its accuracy in classifying diseases.

6. Model Evaluation

After training, the model is evaluated using metrics including:

- **Accuracy:** Measures the overall correctness of the model by calculating the proportion of total predictions that are correct.
- **Precision:** Indicates how many of the samples predicted as a particular class are actually correct, showing the model's reliability in positive predictions.
- **Recall:** Shows how well the model can identify all actual positive cases by measuring the proportion of true positives detected.
- **F1-score:** The harmonic mean of precision and recall, providing a balanced measure when classes are imbalanced.
- **Confusion matrix:** A table that displays true positives, true negatives, false positives, and false negatives to summarize the model's classification performance.

7. Deployment

Finally, the trained CNN model is deployed through a web application. Farmers can capture images of silkworms using a smartphone, upload them, and receive instant disease predictions. The system is designed to be user-friendly, practical, and accessible in rural sericulture environments, facilitating early detection and timely intervention.

CHAPTER 4

SYSTEM DESIGN AND IMPLEMENTATION

The implementation plan for the Smart Sericulture System represents a comprehensive, full-stack deployment strategy across three architectural tiers, designed to seamlessly integrate advanced Convolutional Neural Network (CNN) technology with practical farmer-facing tools. The plan begins with the Flow Chart Introduction, establishing the chart as the essential visual guide that translates the system's requirements into a sequential logic, demonstrating how the farmer's action, such as choosing "Diagnosis" or "Information," dictates the subsequent sequence of processes, decisions, and data handling within the application. The subsequent Implementation involves the Presentation Tier, focusing on the user experience by implementing the Web Interface using HTML, CSS, and JavaScript; this includes building multi-lingual support and robust Image Upload forms to handle the raw input data. This input is then passed to the complex Application Tier, which is centered around the Flask Backend Framework; this tier handles the coordination of the Image Pre-processing (using OpenCV for resizing and normalization), the execution of the AI Service (where the TensorFlow/Keras CNN models perform image classification to return a disease ID and confidence score), and the separate Market Intelligence function implemented via a Beautiful Soup Web Scraper to fetch real-time cocoon prices. Finally, the Data Tier Implementation ensures the system's reliability by mandating the secure storage of the large Trained CNN Models and the construction of the Remedy Database (likely using SQLite/MySQL), which must be populated with structured symptoms, causes, and control measures for every predicted disease, thereby serving as the reliable knowledge repository that transforms a raw AI prediction into an actionable Diagnosis & Remedy Report for the farmer.

4.1 SYSTEM REQUIREMENTS:

4.1.1 Hardware Requirements:

- Processor: Intel Core i5
- Hard Disk: At least 250 GB of storage
- Ram: 8 GB or above
- GPU (Graphics Processing Unit)
- Camera or smartphone.
- 64-bit operating system.

4.1.2 Software Requirements:

- Operating system: windows
- Coding Language: Python
- IDE: Visual Studio Code

4.2 FLOW CHART WITH SYSTEM MODULES:

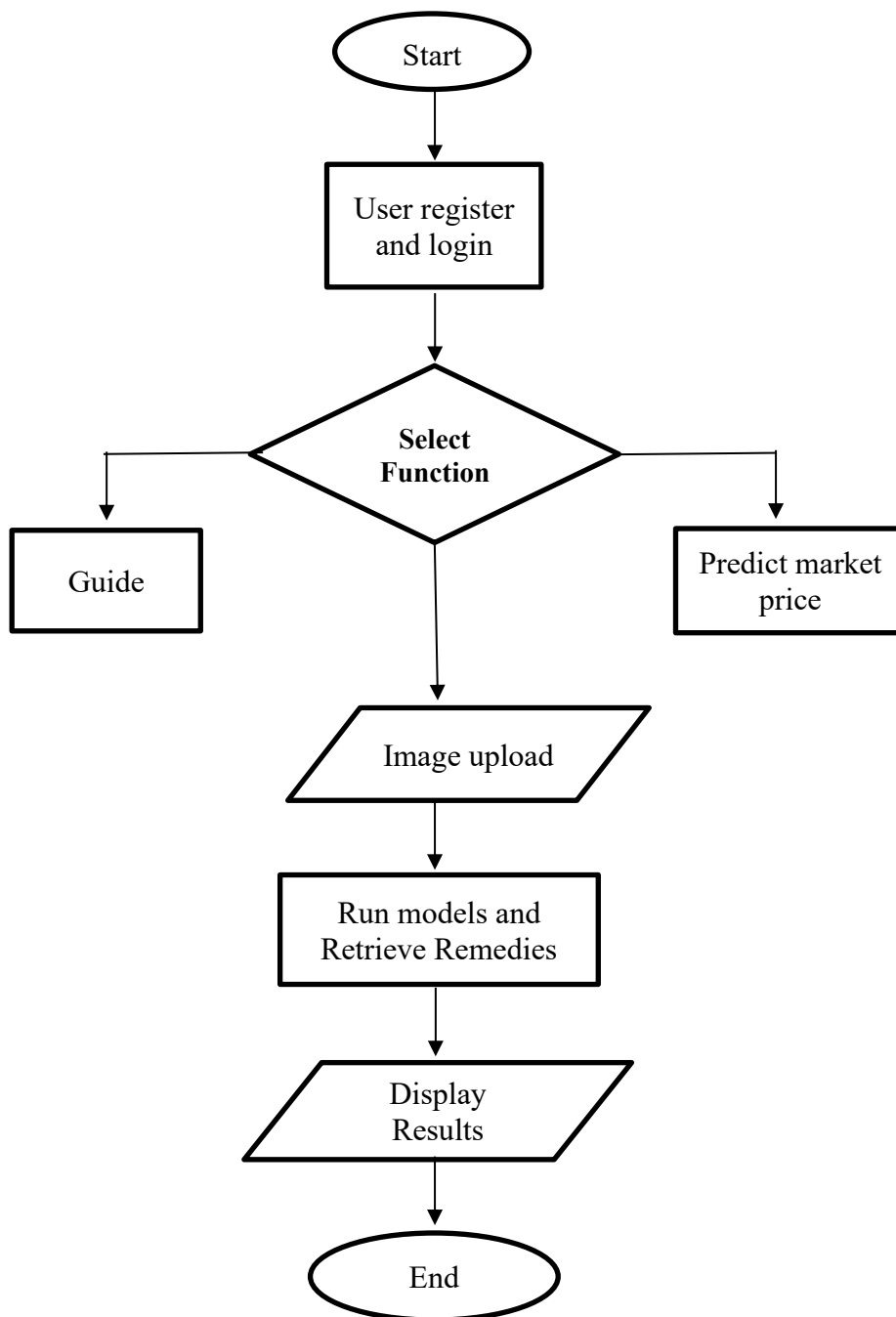


Fig 4.1: Flow chart

Smart Sericulture system operates in three main domains: the User Module (the farmer's actions), the System/Machine Learning Module (the core processing), and the Database/Output Module (data retrieval and presentation).

1. User Module (Input and Decision)

This module represents all the steps taken by the farmer interacting with the application.

- **START (Oval):** The process begins when the farmer accesses the Web Application.
- **User register and login:** farmer will register and login.
- **Select Function? (Diamond):** The farmer makes the first critical decision: choosing between Disease Diagnosis or Information (Market Price/Guide).
- **Image Upload (Parallelogram):** If Diagnosis is chosen, the farmer uploads the image (input) of the suspicious silkworm or leaf.

2. System/Machine Learning Module (Processing and Analysis)

This module handles the core computational tasks, primarily utilizing the CNN models.

- **Pre-process Image:** The system prepares the image by resizing and normalizing it to meet the exact input requirements of the AI model.
- **Run CNN Models (Rectangle):** The pre-processed image is fed into the correct Convolutional Neural Network (CNN) for immediate pattern recognition and disease classification (e.g., Flacherie, Pebrine).
- **Predict Market Prices (Rectangle):** (Parallel to Diagnosis) The system runs the web scraping script to retrieve the latest commercial cocoon prices.

3. Database & Output Module (Data Retrieval and Presentation)

This module provides the practical advice and displays the final results to the user.

- **Retrieve Remedy (Rectangle):** Based on the CNN's diagnosis, the system accesses the database to retrieve the specific causes, symptoms, and control measures for the identified disease.
- **Load Farming Guide (Rectangle):** (Parallel to Diagnosis) The system fetches the structured, week-by-week sericulture guidance for the farmer.
- **Display Results (Parallelogram):** All final information—the prediction, confidence score, remedies, market price, or farming guide—is presented to the farmer on the screen.
- **END (Oval):** The process concludes, having delivered the requested information or diagnosis.

4.3 ALGORITHMS:

- This module utilizes Deep Learning algorithms and various image processing techniques for accurate diagnosis.
- **Convolutional Neural Networks (CNN):** This is the core algorithm used for both Silkworm and Mulberry Leaf Disease Detection.
 - Function: CNNs are specialized in image classification. They automatically learn hierarchical patterns such as colour, texture, and morphological irregularities in the images, bypassing the need for manual feature selection.
 - Architecture: The model architecture includes Convolutional Layers (for feature extraction), Pooling Layers (for dimensionality reduction), and Fully Connected Layers (for final classification).
 - Activation: Re-LU (Rectified Linear Unit) is used as the activation function within the convolutional layers.
 - Output Layer: The final layer uses the Soft-max output layer to predict the probability of the image belonging to one of the disease classes (Healthy, Pebrine, Grasserie, Flacherie, Muscardine, etc.).
- **Training and Optimization Algorithms:**
 - Optimizer: The Adam Optimizer is used to update the model's weights during training.
 - Loss Function: Categorical Cross-Entropy Loss is employed to measure the error between the predicted and actual disease class.
- **Image Pre-processing Techniques:** These methods ensure the image data is clean and uniform before model training or prediction.
 - Resizing & Normalization: All images are resized to a fixed dimension (e.g., 224×224 pixels) and normalized (scaling pixel values to a range like $[0, 1]$) for uniformity and faster convergence.
 - Filtering: Gaussian filtering and histogram equalization are applied to improve contrast and suppress image noise.
 - Data Augmentation: Techniques like rotation, horizontal and vertical flipping, and zooming are used to artificially expand the dataset, which helps reduce overfitting and improves the model's robustness to varying lighting/background conditions.

➤ Database & Output Module (Data Retrieval and Frameworks):

This module relies on specific technologies to gather external data and host the application.

- **Web Scraping (BeautifulSoup4):** The system uses a Web Scraper (specified as price.py in one document) to automatically access the Karnataka Sericulture government website and extract the daily commercial cocoon market prices. This automation is likely implemented using a Python library.
- **Web Application Framework (Flask):** The entire system is built and deployed as a simple web application using the Flask framework (Python backend). This framework is responsible for handling routing, managing user sessions, and processing the prediction requests in real-time.

4.4 SYSTEM DESIGN

4.4.1 Detailed design

The Smart Sericulture system is a full-stack, three-tier web application designed to provide automated disease diagnosis and crucial market intelligence to silk farmers. The core of the system is the integration of Deep Learning models within a user-friendly platform.

1. Presentation Tier: User Interface and Experience (UX)

The Presentation Tier is the user-facing part, ensuring accessibility for all farmers, including those in rural areas.

- **Technology Stack:** Built using standard web technologies: HTML for structure, CSS for visual design, and JavaScript for client-side interactivity and validation.
- **Functionality:**
 - **User Routing:** Acts as the entry point, presenting the farmer with clear choices for Diagnosis, Market Price, or Farming Guide. This aligns with the initial Select Function? decision point in the flow chart.
 - **Input Capture:** Contains a dedicated component for Image Upload, which captures the silkworm or leaf image and sends it as input to the backend server.
 - **Output Rendering:** The final stage, responsible for the Display Output. It cleanly renders the prediction, the confidence score, and the step-by-step remedies in a readable format, often supporting local languages like Kannada for maximum benefit.

2. Application Tier: The System's Core Logic and AI

The Application Tier, running on a server, is the brain of the project. It uses a Python-based backend to manage logic and host the AI models.

- **Backend Framework:** Flask is used as the lightweight web application framework. It manages the server-side requests, handles the incoming image data, and coordinates the entire workflow—from pre-processing to model execution.
- **Image Processing Subsystem:** This subsystem performs the Pre-process Image step. It uses libraries like OpenCV to standardize all input images through:
 - **Resizing:** Scaling images to a fixed dimension (e.g., 224×224 pixels) for model uniformity.
 - **Normalization:** Scaling pixel values to a $[0, 1]$ range to ensure faster and more stable model convergence.
- **Deep Learning Subsystem:** This is the diagnostic engine, built on TensorFlow/Keras.

- Model Hosting: The application loads two key Convolutional Neural Network (CNN) models: one for Silkworm Disease Diagnosis (classifying Pebrine, Grasserie, Flacherie, etc.) and one for Mulberry Leaf Disease Detection. This handles the Run CNN Models steps.
- Prediction: The model executes the image classification algorithm and outputs a probability score for each disease, identifying the most likely affliction.
- Web Scraping Module: This external module executes the Fetch Market Prices function. It uses Python libraries like Beautiful Soup to scrape real-time cocoon price data from the designated government or market websites.

3. Data Tier: Storage and Knowledge Base

The Data Tier is the persistent storage layer, providing essential remedies and guides.

- Model Storage: This consists of the stored trained CNN model files (typically in the .h5 format). These are the weights and configurations the Application Tier loads for immediate prediction.
- Remedy Database: This database (which could be a simple SQLite database or a structured set of JSON files) holds all the non-AI knowledge. It performs the Retrieve Remedy and Load Farming Guide functions.
 - Disease Data: Contains specific symptoms, causes, and detailed control measures mapped to every disease the CNN can detect.
 - Guidance Data: Stores the structured, week-by-week sericulture farming guide.

4.4.2 Data Flow Diagram

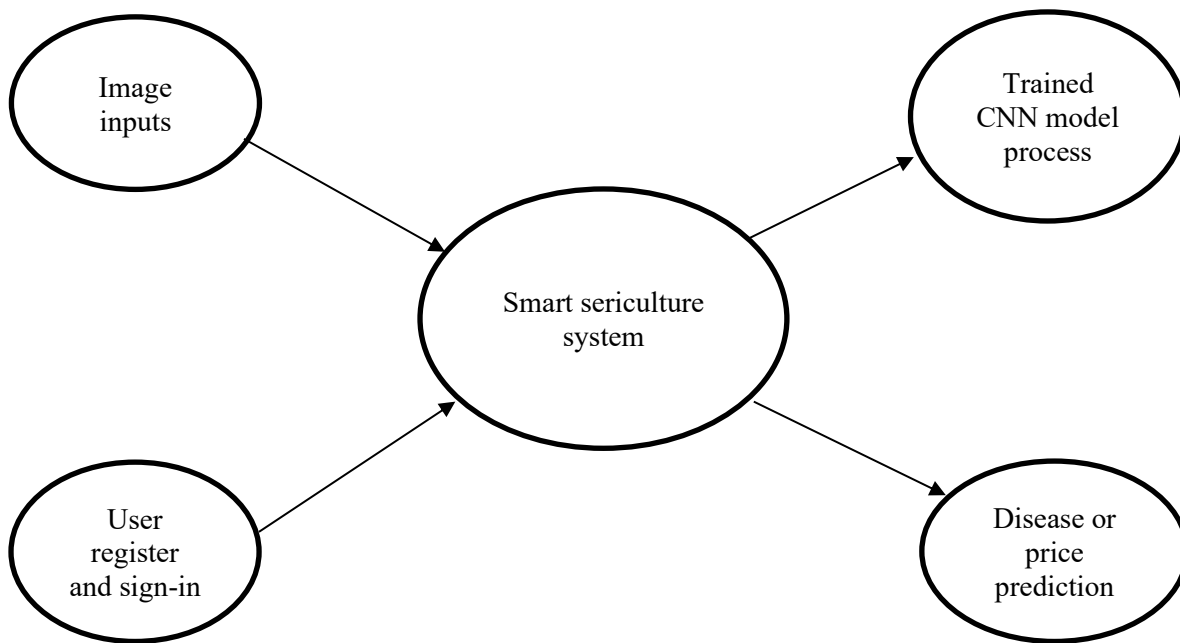


Fig 4.2: Level 0 Data Flow Diagram

Level 0 Data Flow Diagram: A Level 0 DFD (Context Diagram) shows the system as a single process, with all interactions happening at the boundary. The Smart Sericulture System serves as the central hub connecting the farmer to crucial data and AI analysis. The system receives primary inputs from the Farmer (User), including Image Data (of silkworms or leaves) for diagnosis and Function Requests (for market prices or guides). Additionally, it autonomously retrieves Cocoon Price Data from the External Market Website via its scraping module, and it depends on the Trained Model Developer for the initial Trained CNN Models that power the AI. After processing these inputs, the system delivers two main reports back to the farmer: the Diagnosis & Remedy Report (containing the AI prediction and cure steps) and the Market & Guide Report (containing the latest prices and sericulture guidance). This diagram defines the scope of the project by showing all data flowing into and out of the entire application.

4.4.3 Use case diagram

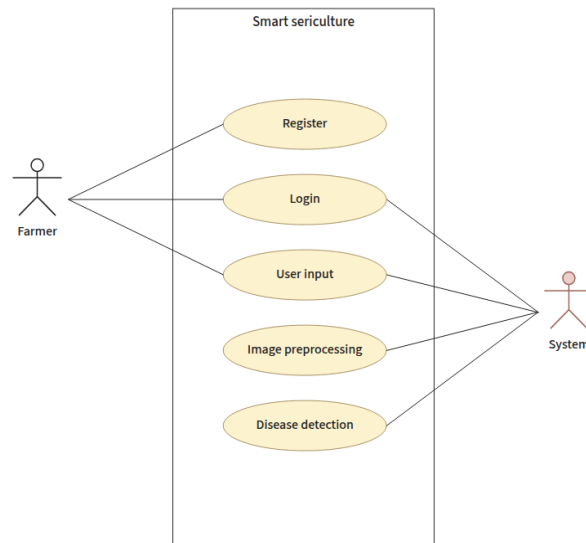


Fig 4.3: Use case diagram

The Use Case Diagram for the Smart Sericulture System provides a complete behavioral overview, defining the entire functional scope of the application solely through the perspective of the core user, the Farmer (Sericulturist). The farmer is the single, central actor who leverages the system's intelligent capabilities to manage their silk production process. The system offers four distinct, user-initiated Use Cases that fulfill the project's goal of automated diagnosis and knowledge transfer. The most critical functions involve the AI Diagnosis based on Convolutional Neural Networks (CNNs): the farmer can initiate the Diagnose Silkworm Disease use case by uploading an image of a silkworm larva to quickly predict one of the five primary disease categories (Pebrine, Grasserie, Flacherie, etc.), or perform the complementary Diagnose Mulberry Leaf Disease use case to check the health of the host plants. Once the AI processing is complete, the farmer then executes the essential View Diagnosis & Remedies use case, which consolidates the prediction, confidence level, and the specific control measures retrieved from the database, delivering actionable advice. Finally, the system supports the farmer's operational knowledge through the View Farming Guide use case, allowing immediate access to the week-by-week instructions and best practices stored in the application's knowledge base, thereby integrating all core functionalities into a single, comprehensive user experience.

4.4.4 Sequence Diagram

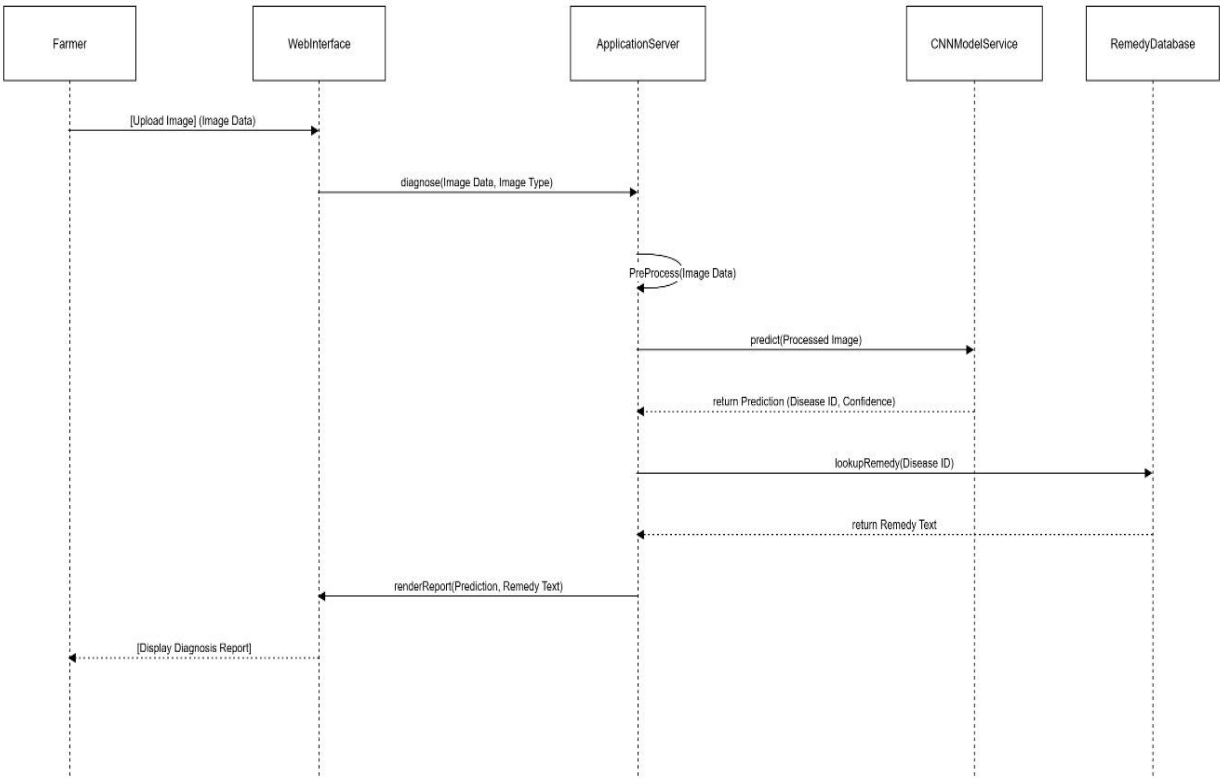


Fig 4.4: Sequence diagram

The Sequence Diagram for the Smart Sericulture System details the exact chronological order of messages exchanged between the Farmer and the system's internal components to successfully perform the Automated Disease Diagnosis. The process begins and ends with the farmer, but the complexity occurs entirely within the server's backend, involving four key lifelines: the user's Web Interface, the coordinating Application Server, the specialized CNN Model Service, and the Remedy Database.

The flow is initiated when the Farmer sends the Image Data of the silkworm or leaf to the Web Interface. This interface then passes the request, along with the image and its type, to the Application Server via a `diagnose ()` call. The server, acting as the orchestrator, first performs a critical self-call to `Preprocess` the image, ensuring it is resized and normalized to meet the AI model's precise requirements. Once cleaned, the server executes the core task by calling the CNN Model Service with the `predict ()` function. This AI service processes the image and returns the Prediction (Disease ID and Confidence Score) back to the Application server.

With the diagnosis secured, the server immediately queries the Remedy Database using the predicted Disease ID via a lookup Remedy () call. The database responds by returning the specific Remedy Text (causes, symptoms, and control measures). Finally, the Application Server compiles all the results—the AI's prediction and the retrieved remedy—into a single report and sends it back to the Web Interface, which then renders and Displays the Diagnosis Report to the Farmer, concluding the entire sequence.

4.4.5 Flow chart

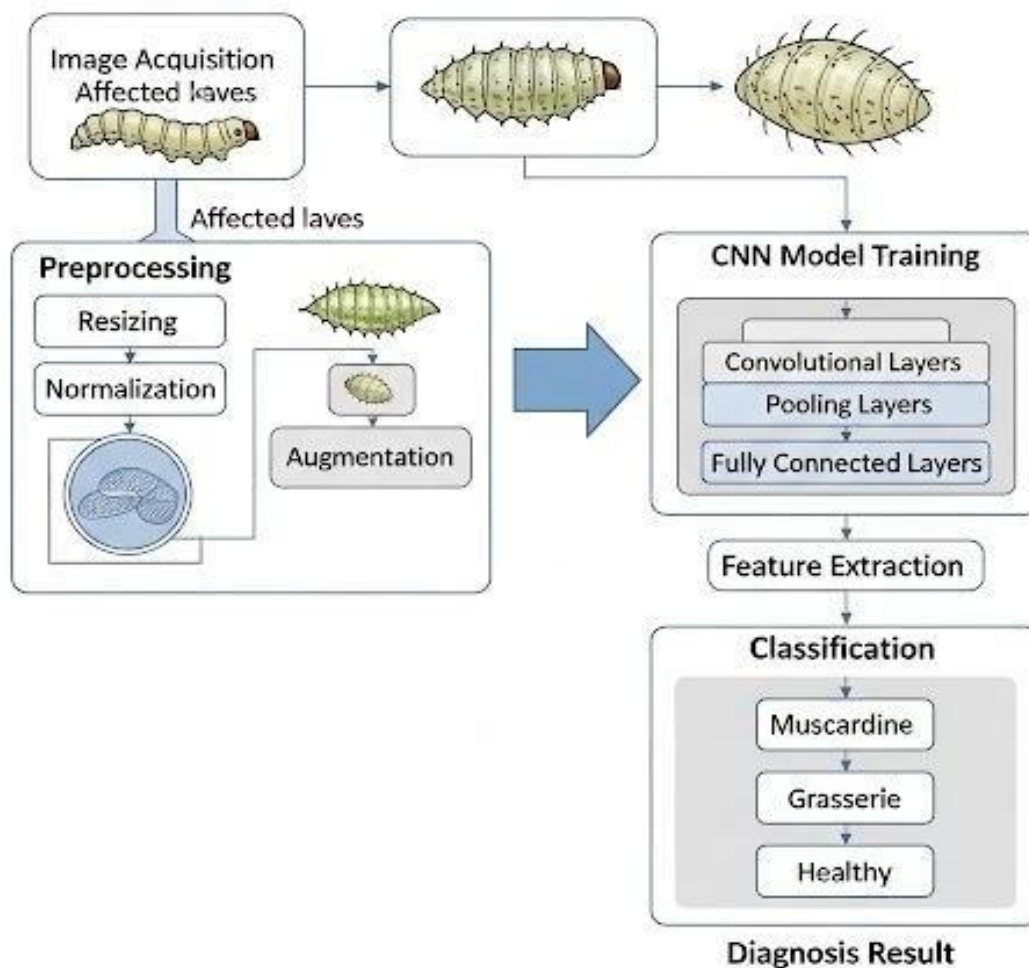


Fig 4.5: Block Diagram

This flowchart illustrates the complete pipeline for diagnosing silkworm larval diseases using Convolutional Neural Networks (CNNs). It shows how raw images of larvae are processed, analyzed, and finally classified into disease categories.

1. Image Acquisition.

In this initial stage, images of affected silkworm larvae are captured using a camera or imaging device.

Purpose: To obtain a dataset containing larvae visually showing symptoms of various diseases.

Ensure images cover different lighting, positions, sizes, and backgrounds.

Output: Raw images of healthy and diseased larvae.

2. Preprocessing

The acquired images undergo several preprocessing steps to prepare them for training the CNN model.

○ Resizing

All images are resized to a uniform dimension (e.g., 224×224 pixels).

Ensures consistency for computational efficiency and CNN input requirements.

○ Normalization

Pixel values are scaled (e.g., 0–1).

Helps stabilize and speed up the learning process.

○ Data Augmentation

Techniques like flipping, rotation, zooming, and shifting are applied. Increases dataset size artificially.

Reduces overfitting and improves model robustness.

Output: A clean, standardized, and expanded set of images suitable for model training.

3. CNN Model Training

A Convolutional Neural Network is trained on the preprocessed images.

a. Convolutional Layers

Extract important features such as shape, texture, color patterns, and disease spots. Detect edges, contours, and fine details.

b. Pooling Layers

Reduce dimensionality. Retain important features while lowering computational cost.

c. Fully Connected Layers

Connect extracted features to possible disease classes. Responsible for final decision-making during classification. Output is a trained CNN model capable of recognizing disease-specific patterns.

4. Feature Extraction

The CNN automatically extracts high-level features from input images. These features represent the unique characteristics of each disease. Output is vector of learned features from the image.

5. Classification

After feature extraction, the CNN model classifies each larval image into one of the following five categories:

1. Grasserie

- A viral disease.
- Larvae appear swollen, shiny, and the body becomes fragile.

2. Flacherie

- A bacterial/viral digestive disorder.
- Larvae become soft, flaccid, and may emit a foul smell.

3. Muscardine.

- A fungal disease.
- Larvae turn stiff and are covered with white/green fungal spores.

4. Pebrine

- A protozoan disease.
- Shows black pepper-like spots on the body (diagnostic hallmark).

5. Healthy

- Normal larva with no visible signs of disease.

CHAPTER 5

RESULTS AND DISCUSSION



Fig 5.1: Home page

This is the homepage of the Smart Sericulture system, featuring a background image of silkworms and a central title. It provides quick navigation through buttons like Home, About Us, Sign In, and Sign Up. The page serves as the main entry point for users to access the platform.



Fig 5.2: Dashboard

This page is the Smart Sericulture dashboard, showing navigation options like silkworm disease prediction, leaf disease prediction, stages information, and current market price. It features a background image of silkworms and a main title welcoming users to the platform. The page serves as a central hub for accessing all major features of the system.



Fig 5.3: Disease Prediction interface

This page is the Silkworm Disease Prediction interface of the Smart Sericulture system. It allows users to upload a silkworm image using the “Choose File” button and click “Predict” to detect possible diseases. The background image and title highlight the focus on silkworm health analysis.



Fig 5.4: Disease Prediction

This page displays the silkworm disease prediction result, showing the uploaded image and the detected disease- Grasseria. It provides clear sections for the causes, symptoms, and control measures of the disease to guide farmers. The page helps users understand the diagnosis and take preventive or corrective actions.



Fig 5.5: Stages of silkworm

This page provides a Worm Rearing Guide, explaining the 2nd stage (2nd instar) of silkworm growth and moulting. It includes details on feeding, moulting duration, and signs such as swollen heads and immobility. The guide helps rearers manage silkworms properly during this developmental stage.

Smart Siriculture			
Home About Us Services Contact Us			
Transaction Date	Language	Source of Report	PDF
05-05-2023	KANNADA	MARKET SECTION	5-5-23.pdf
04-05-2023	KANNADA	MARKET SECTION	4-5-23.pdf
03-05-2023	KANNADA	MARKET SECTION	3-5-23.pdf
02-05-2023	KANNADA	MARKET SECTION	2-5-23.pdf

Fig 5.6: Market Price

This page displays the current market price section of Smart Sericulture, listing transaction dates along with language, report source, and downloadable PDF links. It provides farmers easy access to official market reports in Kannada. The table helps users stay updated on recent silk market price information.

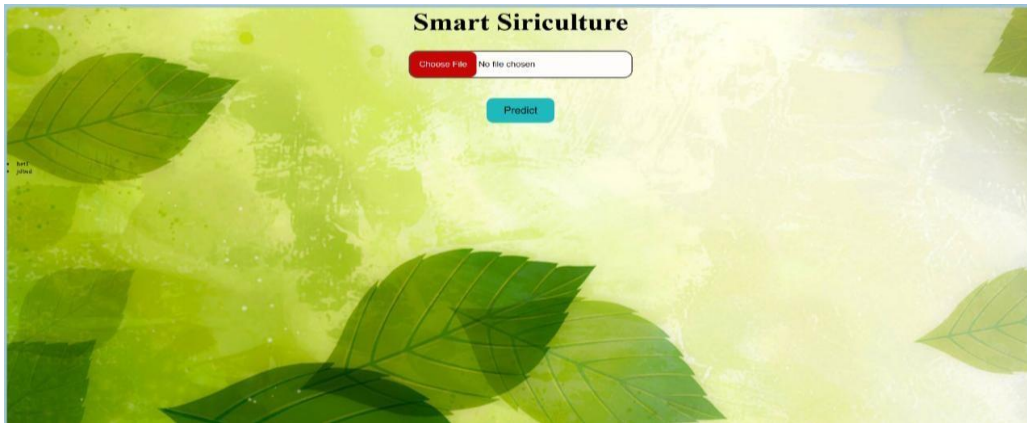


Fig 5.7: Leaf Disease Prediction interface

This page is the Leaf Disease Prediction upload interface, where users can choose a leaf image and click “Predict” to analyze its health. The background theme with leaves represents the focus on mulberry plant monitoring. It serves as the starting point for detecting leaf diseases in the Smart Sericulture system.

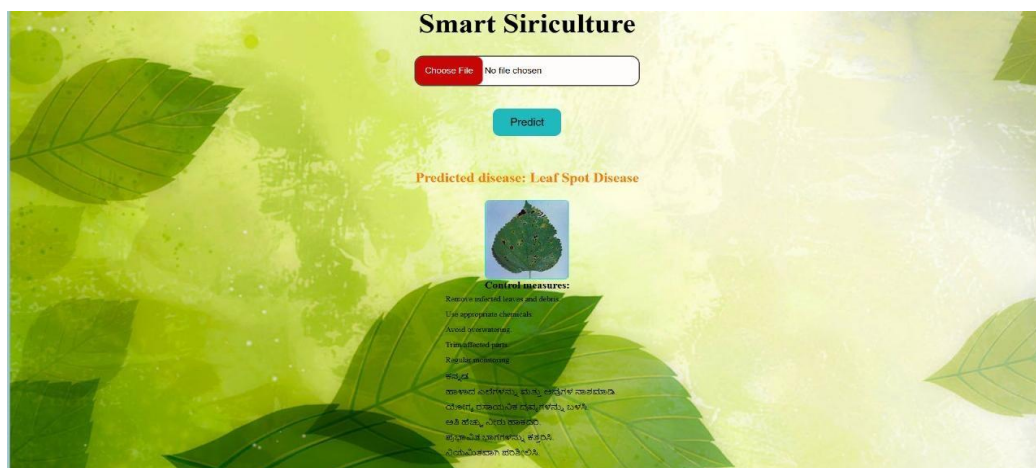


Fig 5.8: Leaf Disease Prediction

This page shows the Leaf Disease Prediction result, where the system identifies the disease as Leaf Spot Disease. It displays the uploaded leaf image along with recommended control measures for managing the infection. The page helps farmers take quick action to protect mulberry plant health.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Conclusion:

The Smart Sericulture system successfully integrates machine learning and image processing to provide an efficient and automated solution for detecting diseases in silkworms and mulberry leaves. By using a Convolutional Neural Network (CNN), the model accurately analyzes uploaded images and identifies common diseases, enabling farmers to take timely preventive and corrective actions. The platform further enhances sericulture management by offering disease details, control measures, rearing guidelines, and market updates in a user-friendly interface. Overall, the project demonstrates how AI-driven tools can significantly improve silkworm health monitoring, reduce crop loss, and promote smarter and more sustainable sericulture practices.

Future scope:

The Smart Sericulture system has immense potential for expansion and improvement. Several enhancements can be integrated into future versions to make the platform more accurate, accessible, and impactful for farmers. The key future scope points are as follows:

1. Integration of Advanced Deep Learning Models

Future work can include more sophisticated architectures such as ResNet, EfficientNet, or Vision Transformers (Vi-T) to improve the accuracy of silkworm and leaf disease classification. These models can handle complex image patterns and enhance prediction precision.

2. Expansion of Disease Categories

The current model can be expanded to detect a wider range of silkworm and mulberry leaf diseases. This will make the system more comprehensive and useful across multiple rearing stages.

3. Real-Time Monitoring Using IoT

By integrating IoT sensors the system can provide real-time environmental monitoring. Instant alerts can be sent to farmers when conditions become unfavorable for silkworm rearing.

4. Mobile Application Development

A dedicated mobile app can make the system more accessible, enabling farmers to upload images, receive predictions, and access guidelines from anywhere.

5. Automated Cocoon Quality Grading

Machine learning can be used to automatically evaluate cocoon quality, helping farmers estimate market value and improve production practices.

6. Yield Prediction and Fertilizer Recommendation

Using machine learning models, the system can be extended to predict yield outcomes and recommend the right fertilizers based on plant health and soil conditions.

7. Cloud-Based Storage and Analytics

Cloud integration can enable large-scale data storage, trend analysis, and faster processing. It will also improve the system's scalability across different regions.

8. Farmer Advisory and Chatbot Support

AI-based chatbots can be included to guide farmers with disease management, feeding schedules, and best practices.

REFERENCES

- [1] Shilpa B. Kodli and Anita, A Novel Approach for Silkworm Disease Detection and Classification by Using CNN and Image Processing Techniques, 2025.
- [2] Suchitha H. S., Anish A. R., Gagandeep J. E., Manoj M., and Manu G., Silkworm Yield Prediction: Exploring Leaf Types and Species, 2025.
- [3] Yutong Liu et al., A Comprehensive Prediction System for Silkworm Acute Toxicity Assessment of Environmental and In-Silico Pesticides, 2024.
- [4] Manju G., Grasserie Disease Detection in Silkworm Using Machine Learning Methods, 2024.
- [5] Raghavendra C. G. and Dharini Raghavan, Grasserie Disease Identification in Bombyx Mori Silkworm Using Ensemble Learning Approach, 2022.
- [6] Jonali Goswami, Nasreen Rasid, and D. K. Gogoi, Development of a Muga Disease Early Warning System A Mobile-Based Service for Seri Farmers, 2021.
- [7] Hui Xiang, Xiaojingnd Muwang Li, The Evolutionary Road frond Moth to Domestic Silkworm." 2018.
- [8] Sandeep Kumar S. and Usha Divakarla, Mulberry Leaves Diseases and Disease Identification Techniques, Proc. Int. Conf. Integrated Intelligence and Communication Systems (ICIICS), IEEE, 2023.
- [9] Shilpashree P. S., Sampath G. D., Pavan kumara swamy H. S., Sagar B. E., and Chaithra S. K., Classification of Healthy and Diseased Mul-berry Leaves Using Machine Learning, Proc. Int. Conf. Smart Systems for Applications in Electrical Sciences (ICSSES), IEEE,

